

Can AI Help Screen Ac-225 Production Plans Faster?

A physics-guided project testing whether many possible irradiation and cooling schedules can be screened quickly, then checked with trusted nuclear calculations.



What I Have Completed

- Built a reduced five-isotope physics model
- Trained the V3 LSTM for 6,000 epochs
- Froze the model before new comparisons
- Checked speed against exact and Radau solvers
- Generated 1,400 training examples with Radau
- Compared it with a similarly sized non-recurrent model
- Audited 10,000 simplified scenarios
- Preserved the failed Berkeley transfer test

What Must Happen Next

- Decide exactly what a real researcher needs the model to choose.
- Obtain realistic neutron, beam-timing, target-geometry, and uncertainty data.
- Have an expert check the tracked isotopes and impurity goal.
- Generate richer training examples with trusted nuclear software.
- Set aside an unseen benchmark before retraining once.
- Recheck every suggested finalist with trusted software.

Current evidence: On 10,000 simplified computer scenarios, V3 was 84.9 times faster than the general Radau solver but 56.7 times slower than the exact calculation for the same reduced equations. It is a prototype for screening first and verifying second; it has not yet shown that real Ac-225 production is faster, cheaper, or experimentally validated.